Annual Global Update

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1 Introduction

This report provides the global status of the use of spectrum in the 410 MHz and 450 MHz bands. The information is gathered from operators and regulators worldwide by 450 MHz Alliance. The 450 MHz Alliance is an interest group fostering deployments in the 380 to 512 MHz band. Primary activities of the 450 MHz Alliance include advocacy for evolution of standards, technical education, advocacy, regulatory affairs, global cooperation's, system and device availability and cooperation to develop new features and services.

Worldwide, there are currently twenty-six LTE operations with commercial traffic and another four operations performing trials. CDMA networks are still the dominant technology in the 410 MHz and 450 MHz and regulatory work for converting technology specific licenses is an obstacle in the progress of evolving to 4G. Europe is currently leading the conversion from CDMA to LTE with Asia and South America in progress. North America and Australia have been dominated by narrowband systems in the 400 MHz bands, but are currently review the possibility of introducing local permits for industries in the bands. Most licenses today are nationwide, but a trend with local licenses for industry application are growing. The aim of this report is to provide operators with a simple introduction to the 450 MHz ecosystem for their business and to show business opportunities to the suppliers who provide equipment to operators and end-users.

2 Technology Overview

LTE in 450 MHz bands is a part of the 3GPP standard for 4G/LTE. With advanced data and voice capabilities, flexibility and a seamless migration path to next generation technologies (typical 5G), LTE has become a leading wireless technology for delivering voice and broadband data to densely populated urban areas as well as rural and remote regions in developed and developing markets. LTE450 – or LTE in the 450 MHz frequency bands – is a mature and robust technology for cost-effective provisioning of both basic and advanced voice and broadband data services across regions with low population densities or difficult terrain due to the favorable propagation characteristics of the lower frequency band combined with a robust and advanced 4G LTE technology. The unique characteristics due to low spectrum give a superior coverage compared to any other spectrum bands standardized by 3GPP and ITU. The typical coverage area for different spectrums can be seem in the figure 1, below.

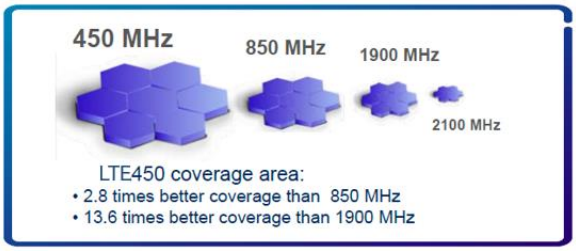


Figure 1. Coverage area for different spectrums

Being part of the 4G/LTE and 3GPP ecosystem brings the advantages of a global technology evolution for connectivity. The fundamental broadband services have evolved to support narrowband and lower power connectivity as well and voice and critical services with access priority and quality of service support. Low power and narrowband support for LTE-M (eMTC) and NB-IoT are available from several suppliers. The path to 5G are in progress, the 450 bands 31 and 72 are expected to complete the 3GPP RAN process by end of 2023 beginning 2024 and will be included in 3GPP release 18. Evolution to 5G follows the general implementation of bandwidth support

for 3 MHz and above, since these smaller bandwidths are applicable in the 400 MHz bands. The current view is that both standalone and non-standalone versions are required to be supported and that dynamic spectrum sharing would be essential for some implementations.

3 Market Overview

The major development in the last twelve months has been the network deployment in Germany, start of network deployment in Poland and Ireland, trial in Malaysia, as well as the progress of spectrum allocation in the Kingdom of Saudi Arabia and the consultation in South Africa. Most existing allocations of spectrum in the 400 MHz bands are still not technology neutral and require regulatory updates or changes to the allocation and/or licenses to be able to implement LTE by the local operators. The trend is that the 400 MHz spectrum bands are more and more being allocated to private networks typical Utilities, Public Safety and Transport. The advantage is the predictability of the operation, it will be stable and controlled since the network is deployed with a dedicated business as a base and customized to the requirements of the user. Due to the conversion of the business model, the number of operators is likely to decrease temporarily before new allocations have been awarded. There are more than twenty networks actively investing in LTE globally.

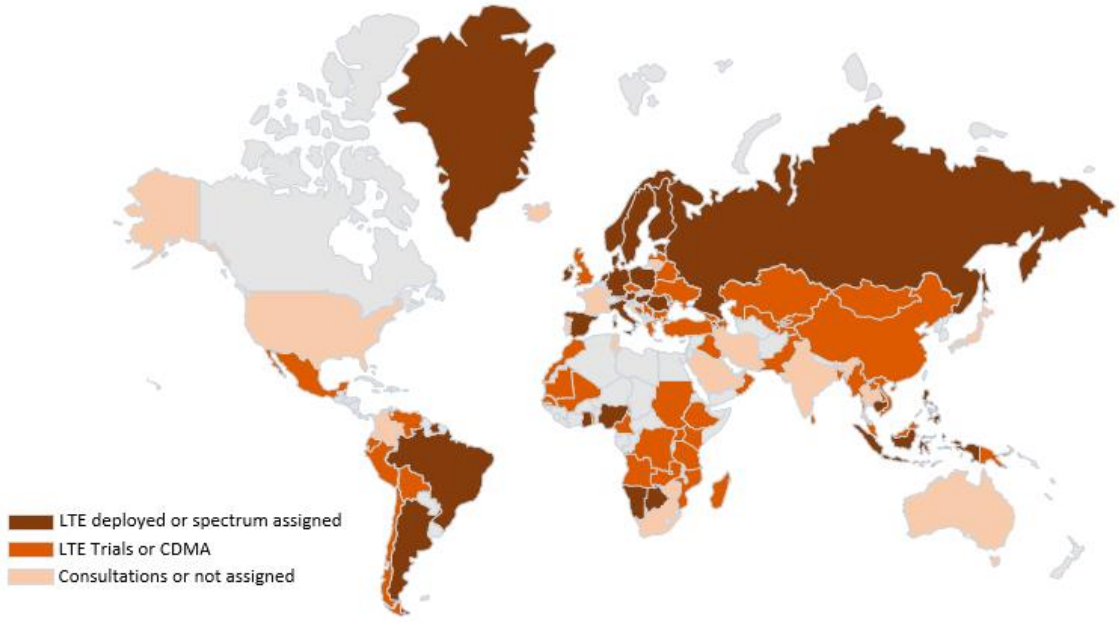


Figure 2. The world map of 380 MHz, 410 MHz and 450 MHz deployment.

Spectrum in the 400 MHz bands is today already allocated or in consultation in countries representing more than 6.7 billion people or 86 percent of the world population, which is a small increase since last year. Networks are being or are already deployed or in trial in countries representing more than 4.0 billion people or 51 percent of the world population, which is a small decrease since last year mainly due to shift in license ownership and changes of technology from CDMA to LTE.

3.1 Applications

Network deployments in the 400 MHz bands are suitable for many applications both as the primary carrier as well as backup service. The main applications implementing 400 MHz bands can be divided into four main segment Utility, Transport, Public Safety networks and Rural connectivity.

Connectivity to support applications and services for **utility** distribution network evolves to require private and dedicated connectivity network due to them supplying critical function in society. This includes connectivity to support smart grid, smart meters, smart city, local production, electrical vehicle charging stations and devices for the staff. Security of the utilities production, distribution, and delivery of services to their customers is the reason to build dedicated networks. The 400 MHz bands fit for providing the coverage required for these applications and can also cater for the capacity need. The organizations UTC (North America), EUTC (Europe), UTCAL (Latin America), UTC Africa and UTC Asia have extensive information regarding Utility application.

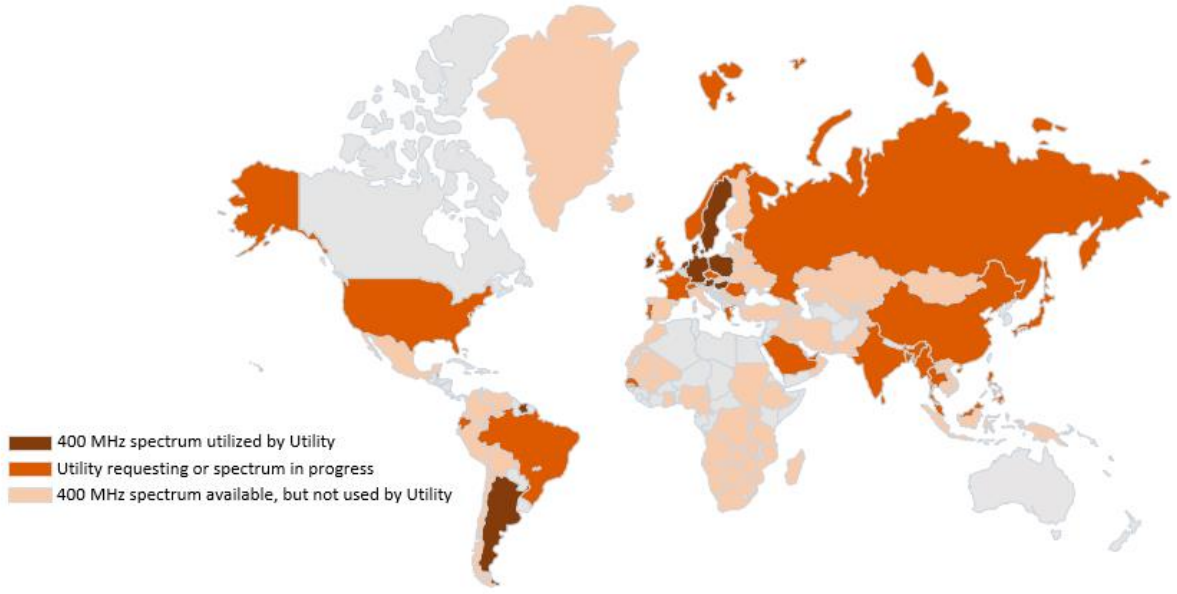
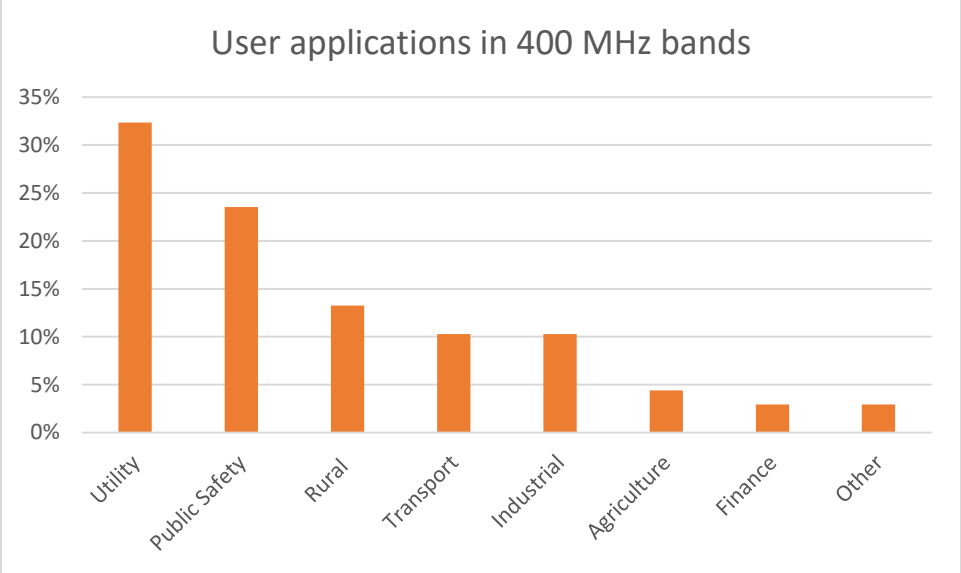


Figure 2. The world map of Utilities interest in the 400 MHz bands.

Public safety would use the 400 MHz bands as part of their connectivity solution either as primary or secondary carriers. Typical service would be broadband and machine to machine, but not limited to these. Networks are likely to have a nationwide coverage but could also be suitable for mobile unit with instant wide range coverage. TCCA provides additional information regarding the requirements for public safety application and services.

Applications for **transport** requires coverage over large range of areas, this includes railways, road service, logistics, mining, shipping, timber industry and agriculture. The solutions can vary from simple monitoring to remote control and assisted automation. The 400 MHz spectrum can provide a secure and continues connectivity for many of the applications and with complementary higher spectrum, today supported by most chipsets, be a seamless part of the overall solution. Like the utility network security of the information and the control of the network drives the requirements for a dedicate and private network.

Rural connectivity includes connectivity for rural school, health care centers, rural ATMs and bank offices broadband also including services for unconnected or poorly connected consumers. This segment enables services that are critical for a modern society.



Graph 1. The global implementation per application.

The 400 MHz bands provide a service for these applications with an attractive business model. This is due to the capacity and coverage possibilities, which enable the desired traffic pattern with an affordable investment.

3.2 Africa

Listed below is a short status update per country for the most active countries in the last twelve to eighteen months.

- South Africa regulator have an ongoing consultation in the 450-470 MHz band.
- The regulator in Tunisia have initiated a consultation including 410-430 MHz and 450-470 MHz.
- There has been a general spectrum consultation Morocco including the 400 MHz bands.
- In Kenya spectrum in the 400 MHz have been reserved for LTE.
- Open Sky Services (MPS) holds the band 31 in Nigeria and have an ongoing procurement of system.
- The regulator in Uganda is looking at awarding the 380 MHz band for PPDR services.
- The band 31 license in Senegal is being evaluated for different industrial verticals.
- Telecom Namibia holds the band 31 license.
- The 450 MHz spectrum in Angola is in progress to evolve from CDMA to LTE by Angola Telecom.
- Botswana regulator BORCA has announced auction of the 450 – 470 MHz band for broadband use.

3.3 Americas

Listed below is a short status update per country for the most active countries in the last twelve to eighteen months.

- The utilities in the USA are seeking opportunities to obtain their own spectrum, mainly to be independent from third parties and to be in control of the connectivity. Both spectrum in the 410 MHz and in 380 MHz is being evaluated.
- Band 31 in Brazil has been taken back to the regulator for the two small licenses holders of the totally four spectrum owners due to lack of deployment. A number of Utilities in Brazil are conducting trials in the 410 MHz band with the goal of being awarded dedicated licenses for the industry.
- Colombia allocated 380 MHz band for PPDR and have had a consultation regarding the spectrum in the 450 MHz band.
- Alvis is the first former CDMA operator in Argentina evolving to LTE. The deployment includes both band 31 and band 87 with focus on, but not limited to, rural areas and agriculture.
- Suriname has a deployed network in the 450 MHz band for IoT services.
- Telmex holds a 450 MHz license in Mexico currently operational on CDMA and are evaluating to convert to LTE.
- In Bolivia the utility evaluates the possibility of using the 450 MHz spectrum.

3.4 Asia and Oceania

Listed below is a short status update per country for the most active countries in the last twelve to eighteen months.

- India has had several ongoing investigations for dedicated spectrum for different segments and verticals. 450 MHz has been included in proposals for railway and critical communication.
- China Unicom holds the band 31 license in China. No official statement has been made with regard to services to be delivered using the spectrum at this time. It has been evaluated for power grid and meters as well as railway communication.
- The license holder of band 31 in the Philippines has changed ownership. The new owner has not officially stated the plans for the operation.
- The band 31 and 72 license in Indonesia were taken back by the regulator and have been awarded to the government owned Bakrie Telecom (Btel). The expectation is that the operation will a dedicated network for governmental systems.
- Band 31 is available in Vietnam. Previous license holder was EVN Telecom.
- Malaysia Telecom has held the band 31 license for many years with limited progress. With the growing need for connectivity, the utility TNB in Malaysia have been reviewing the possibility to use the spectrum.
- The Telecom Regulator in Pakistan has had consulted in order to evaluate the interest in evolving the current CDMA license to a technology neutral or LTE license.
- The Telekom Regulator in Bahrain have awarded EWA a license in band 87.

- The 410 MHz and 450 MHz bands are part of the Saudi Arabia extensive spectrum plan and will hopefully become available during 2022 for deployments.
- The Cambodia regulator has awarded 60 MHz from 390-450 MHz to Titan. The focus is to provide broadband services.
- Singapore have awarded 450 MHz spectrum.
- Dialog holds the 450 MHz license in Sri Lanka and are looking to evolve to LTE.
- Thailand is reviewing the use of 450 MHz to 470 MHz-band.
- Telecom Armenia holds the 450 MHz license in Armenia and provides a broadband service.

3.5 Europe

Listed below is a short status update per country for the most active countries in the last twelve to eighteen months.

- Elisa holds the band 31 license in Finland, but there is no official target segment stated at this time.
- ice holds the band 31 license in Norway which is mainly used for consumer broadband in rural areas typical second homes. A license for band 87 has been out for consultation with limited interest from the main operators, other industries were not included in the consultation.
- Teracom holds the band 31 license in Sweden. It serves broadband for consumers, utilities and mix of business solutions for typical rural areas.
- Cibicom holds the band 31 license in Denmark with the initial focus for the IoT, but no official product launch has been registered at this time. The Danish police holds the license in band 87 and are evaluating the options to progress.
- ESB holds the band 87 license in Ireland and is currently in the procurement process for network infrastructure and devices. The main initial use case is to deploy connectivity for smart grid.
- PGE holds the band 31 license in Poland and are in the planning phase for their deployment. Polkomtel holds the band 87 license and working on upgrading from CDMA to LTE.
- Utility Connect holds the band 31 license in the Netherlands which is mainly used for smart meters and smart grid connectivity.
- 450connect holds the band 72 license in Germany. The operator has started deployment of the network and service. The network will be deployed with a focus on emergency voice communication, smart meters, smart grid and communication of other critical infrastructures.
- ArgoNET is holding the band 72 spectrum licenses in Austria. Multiple regional CDMA and LTE networks have been deployed to address various utility applications.
- SIRDEE holds the band 31 license in Spain and have given Telefonica the task to build and operate a public safety network.
- EDP have had trial network in Portugal. However, the commercial terms have not been at a level enabling the benefits of the deployment, so the process to award the license is ongoing.
- Tele2 Russia, part of Rostelecom, holds the band 31 license in Russia. The network provides services for Public safety staff, industries and transport.

- MVM NET holds the band 31 license in Hungary with primary service for smart city, smart grid and smart meters.
- Greece have allocated a license in band 87.
- Italy have allocated the 450 MHz spectrum for nomadic use as part of a tactical communication system.
- UK have had a consultation regarding 410 and 450 MHz spectrum and have proposed to progress with an allocation for the utilities within medium term time frame.
- France have done a consultation and there is an interest to move forward with an allocation for the band, but the path have not been set.
- Georgia have started to review the possibility of converting the license to technology neutral.

4 Overview of Spectrum

The LTE bands available in the 380 MHz to 512 MHz are currently:

- Band 31 (450 MHz)
- Band 72 (450 MHz)
- Band 73 (450 MHz), not in use
- Band 87 (410 MHz)
- Band 88 (410 MHz)

The 5G bands in progress for 3GPP release 18 in the 380 MHz to 512 MHz are currently:

- Band n31 (450 MHz)
- Band n72 (450 MHz)

All bands are standardized to support LTE, LTE-M and LTE-NB (NB-IoT). Ongoing strategic work with TCCA, EUTC, UTCAL, UTC and other parties in the ecosystem are progressing to identify spectrum allocation in the 380 MHz and 470 MHz bands. During the last twelve months several enquiries have been made in regard to the evolution and standardizing bands for 5G, current plan is that band n31 and n72 will be part of 3GPP release 18 and band n87 and n88 will be part of release 19.

5 Global spectrum allocations

450 MHz provides a more or less global footprint, and the 410 MHz footprint is increasing rapidly. The number of commercial operations is fairly limited in relation to the global mobile industry. This is due to having only one operator per country. The spectrum attracts interest from many segments and the introduction of LTE has made it easier to deploy networks. CDMA licenses are gradually being converted and narrowband systems that have been widely used in this spectrum are being evolved to commercial mobile networks and to more capable and secure private networks which enable additional bandwidth to be available. The global trend of allocating spectrum for private and dedicated networks drives regulators to find fitting frequency ranges these varies a lot pending on the region.

The global variation of spectrum allocation for dedicated or private network can be seen in the table below.

Spectrum / Region	410 MHz 450 MHz	700 MHz	800 MHz 850 MHz 900 MHz	1600 MHz 1800 MHz	2100 MHz 2300 MHz	2500 MHz 2600 MHz	3500 MHz 3700 MHz	4900 MHz	2nd lease
Africa	B31/72/87	B28/68							Yes
Asia	B31/72/87		B26/27			B41		n79	Yes
Europe	B31/72/87	B28/68		B3	B40 n40	B38 n38/n90	B42/43 n77/78		Yes
Middle East	B31/72/87				B1 n65		B43 n77/78		
North America	B31/72/87	B14	B8/106	B54			B48		Yes
Oceania Pacific	B31/72/87		B26/27					n79	Yes
South America	B31/72/87				B40 n40	B38 n38	B42/43 n77/78		Yes

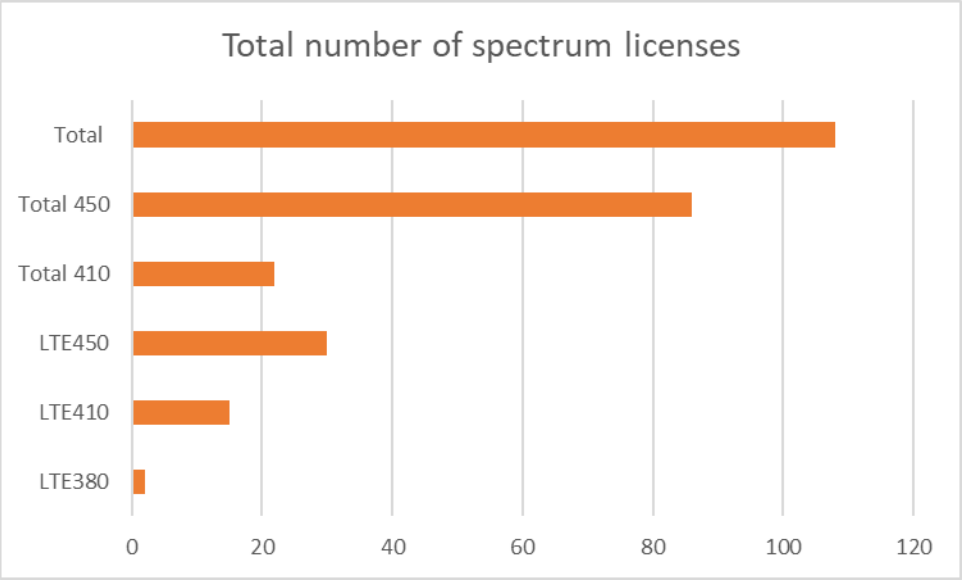
Table 1. Private network spectrum allocations per region

The 450 MHz bands are today the only common band for all regions in the world. The bands that are not used by commercial operators are 410, 450, 700 (B68), 850 (B27), 900 (B106) and 4900 (n79) MHz.

5.1 410 MHz and 450 MHz spectrum allocations listed by 450 MHz Alliance

The annual global spectrum update by 450 MHz Alliance collects information from regulators, operators and suppliers. The following information has been obtained from the responses of 72 regulators. In addition to the regulators, a number of operators, suppliers and industry organizations have made their contributions, which makes the total number of countries with confirmed status 118.

The total number of licenses globally is 113. The progress varies from consultation to deployed and operational networks. Detailed information can be found in the table below.



Graph 2. Number of licenses globally for 380 MHz, 410 MHz and 450 MHz.

Progress of deployments in these spectrum bands is controlled by the individual country. Currently there is no single stakeholder that drives the global progress of evolving or deploying the bands. Below, are the tables showing the status per region globally.

	Deployments	LTE Deployments	Consultation	Not in use
Africa	22	6	4	2
Americas	10	5	6	3
Asia and Oceania	18	3	9	4
Europe	25	16	7	13
Total	75	30	26	22

Table 1. Number of countries per region and globally with 380 MHz, 410 MHz and 450 MHz.

5.2 List of spectrum allocations in Africa

The total amount of networks is twenty-two (22), six (6) of these are LTE. The list below shows the networks reported.

Country	Spectrum	Operator	Technology	Description
Angola	450	Angola Telecom	CDMA	
Botswana	410	Police	LTE	PPDR
Cameroon	450	CAMTEL	CDMA	
DR Congo	C	Tatem Telecom	CDMA	
Ethiopia	450	Ethio Telecom	CDMA	
Gambia	450	Gambia Telecommunications Company Limited	CDMA	

Ghana	B31		LTE	
Madagascar	450	Telecom Malagasy	CDMA	
Malawi	450	Malawi Telecommunications Limited (MTL)	CDMA	
Mali	450	Sotelma	CDMA	
Mauritania	450	Mauritel	CDMA	
Morocco	450	Marco Telecom	CDMA	
Mozambique	450	Telecomunicações de Mozambique	CDMA	
Namibia	450	Telecom Namibia	LTE	
Nigeria	B31/B87	Open Sky Services (MPS)	LTE	
Senegal	450	Sonatel	CDMA	
South Africa	B31/B72		LTE	Consultation
Sudan	450	Canar Telecom	CDMA	
Tanzania	C	Benson Informatics Limited, Tanzania Telecommunications Company Ltd. (TTCL)	CDMA	
Togo	450	Togo Telecom	CDMA	
Uganda	A	Uganda Telecom	CDMA	
Zambia	450	ZAMTEL	CDMA	
Zimbabwe	450		LTE	Open for consultation

Table 2. List of countries in Africa and deployment status.

5.3 List of spectrum allocations in Americas

The total amount of networks is ten (10), five (5) of these are LTE. The list below shows the networks reported.

Country	Spectrum	Operator	Technology	Description
Argentina	B31/B87	Alvis, Several local operators	LTE/CDMA	
Bolivia	450	Startel S.A.	CDMA	
Brazil	B31/B87	Claro, Oi, Vivo, TIM	LTE	Band 87 in consultation
Chile	450	Mobilink	CDMA	
Colombia	B31/B87 and 380		LTE	LTE in 380 MHz allocated, 450 MHz and 410 MHz in consultation
Ecuador	450	Corporación Nacional de Telecomunicaciones (CNT), Etapa Cuenca	CDMA	
Mexico	A/C	Telmex	CDMA	
Peru	450	Movistar	CDMA	
Suriname	450	Telesur	LTE	
U.S.	410		LTE	Consultation

Venezuela	450	CANTV Movilnet	CDMA	

Table 3. List of countries in Americas and deployment status.

5.4 List of spectrum allocations in Asia and Oceania

The total amount of networks is eighteen (18), three (3) of these are LTE. The list below shows the networks reported.

Country	Spectrum	Operator	Technology	Description
Australia	410/450		LTE	Consultation
Azerbaijan	450	Catel (brand FONEX)	CDMA	
Bahrain	410	Utility	LTE	
Bangladesh	C/B31/B72	National TeleCom Ltd (NTC), Peoples Telecommunication & Information Services	CDMA	
Cambodia	B87/B88	Craig Wireless	LTE	
China	A	China Unicom	CDMA	
India	410/450		LTE	Consultation
Indonesia	B31/B72	Net1 Indonesia	LTE	
Iraq	450	Ariafon, Moutiny Wireless Telecommunications	CDMA	
Japan	450		LTE-M/NB	Narrowband solutions
Kazakhstan	450	JSC Kazakhtelecom	CDMA	
Kyrgyzstan	450	Kyrgyz Telecom	CDMA	
Laos	450	Lao Telecommunications	CDMA	
Malaysia	B31	TNB	LTE	Waiting regulatory approval
Mongolia	A	G-mobile	CDMA	
Myanmar	A	Myanmar Post and Telecommunications (MPT)	CDMA	
Oman	450	Oman Telecommunications Company	CDMA	
Pakistan	B31/B72	Pakistan Telecommunications	CDMA	
Pakistan	B31/B72	TeleCard, Worldcall	CDMA	
Papua New Guinea	450		CDMA	Available
Philippines	B31	Broadband Everywhere	LTE	
Saudi Arabia	B31/B87		LTE	Consultation
Singapore	450	Unknown	LTE	
Sri Lanka	450	Dialog	CDMA	Conversion to LTE in progress
Tajikistan	450	JSC Tajiktelecom	CDMA	
Thailand	450		LTE	Consultation

United Arab Emirates	410		LTE	Consultation
Uzbekistan	450	Uzbektelecom	CDMA	
Vietnam	B31		LTE	Available

Table 4. List of countries in Asia and Oceania and deployment status.

5.5 List of spectrum allocations in Europe

The total amount of networks is twenty-five (25), sixteen (16) of these are LTE. The list below shows the networks reported.

Country	Spectrum	Operator	Technology	Description
Armenia	B31	Telecom Armenia	LTE	
Austria	B72	ArgoNET	LTE	
Belarus	A	BelCel JV	CDMA	
Bosnia & Herzegovina	410	Unknown	CDMA	
Bulgaria	450	Bulgarian Telecommunications Company (BTC)	CDMA	
Czech Republic	B72/B87		LTE	Not assigned
Denmark	B31/B87	Cibicom, Police Denmark	LTE	
Estonia	B31/B87		LTE	Not assigned
Finland	B31	Elisa	LTE	
France	B31/B72/87		LTE	Consultation
Georgia	A	Magticom	CDMA	Consultation for LTE
Germany	B72	450connect	LTE	
Greece	B87		LTE	
Hungary	B31	MVM Net	LTE	
Iceland	450			Available
Ireland	B87	ESB Networks	LTE	
Italy	B72	Nomadic use for PS	LTE	
Latvia	A	Triatel	CDMA	
Lithuania	B31		LTE	Available
Moldova	450	Unite	CDMA	
Netherlands	A/B72	Utility Connect	CDMA/LTE	
Norway	B31	Ice	LTE	
Poland	B31/B87	PGE Systemy, Polkomtel	LTE, CDMA	
Portugal	B31		LTE	Trial done by EDP
Romania	B31	Telemobil	LTE	
Russia	B31/B87	Tele2 Russia (Rostelecom)	LTE	B87 in consultation
Serbia	450	Telekom Srbija	CDMA	

Slovakia	450		LTE	Auction pending
Slovenia	B31/72/87		LTE	Consultation
Spain	B72	MoI/SIRDEE	LTE	
Sweden	B31	Teracom	LTE	
U.K.	410	Arqiva		Consultation
Ukraine	A		CDMA	Consultation to be converted

Table 5. List of countries in Europe and deployment status.

6 Auction statistics

The number of auctions enabling the evolution of business to LTE within the 400 MHz have been about ten during the last couple years. The statistics from these auctions give a clear trend: License periods are being extended from between ten and fifteen years to a period of twenty to twenty-five years, the average license duration today is above seventeen years. Auction prices based on price per MHz, population and year have been adapted to services provided by the license holder and is on average 0.31 Eurocent/MHz/Pop/year. It ranges from 0.14 to 0.73 Eurocent/MHz/Pop/year pending conditions of the license. Based on GDP the average is 0.00060 Eurocent/MHz/GDP/year. The majority of the licenses, more than seventy percent, are within forty percent of the average price point.

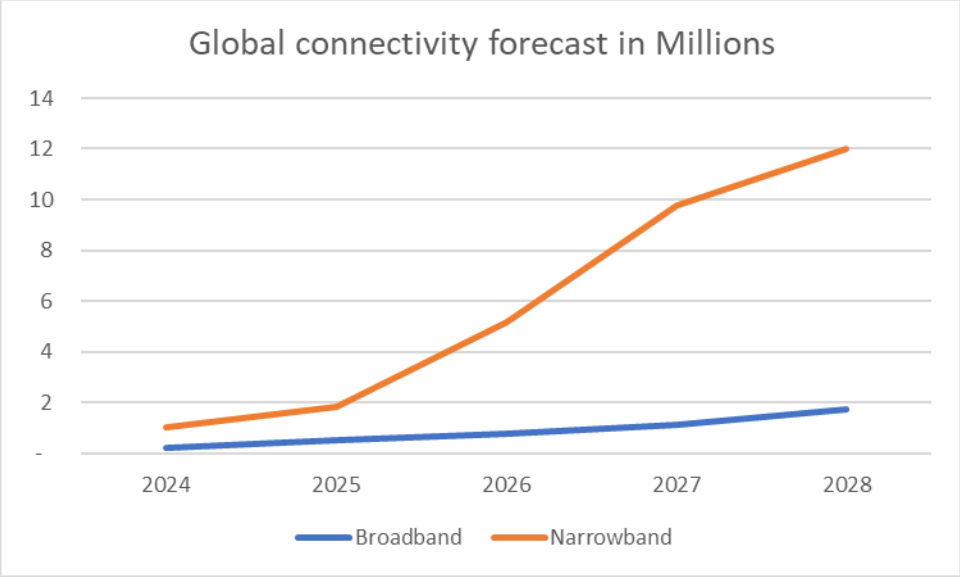
7 General outlook

The evolvement towards LTE of the spectrum in 380 MHz to 512 MHz is expected to increase. Bandwidth and coverage requirements for private and closed networks as well as public networks for rural communication are getting more interest and with this also an increased interest for the lower spectrum bands. With only twenty active operation and spectrum allocated in over seventy countries the expansion can be very fast. The ongoing consultations and the interest from government and companies are likely to drive the ecosystem even faster.

The estimation for connections for the coming years considering the currently know operations in progress are listed in the table below.

	2024	2025	2026	2027	2028
Broadband	223 100	523 000	788 000	1 153 000	1 705 000
Narrowband	998 000	1 811 000	5 146 000	9 785 000	12 005 000

Table 6. Global forecast of connections for services in the 380 to 470 MHz.



Graph 3. Global forecast of connections for services in the 380 to 470 MHz in millions of units.

Broadband includes all connections with a higher category than Cat. 1 and Narrowband includes Cat.1, LTE-M and NB-IoT.